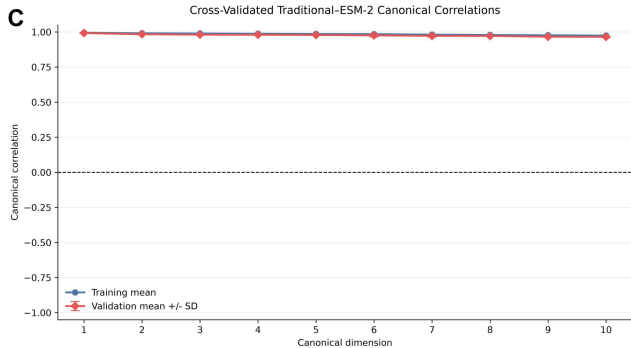
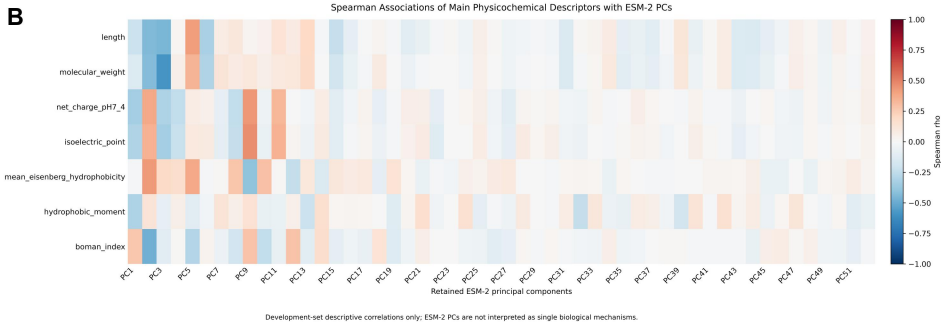
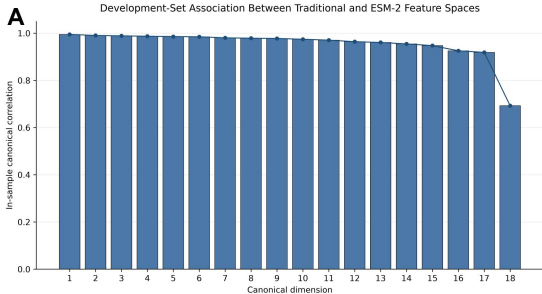
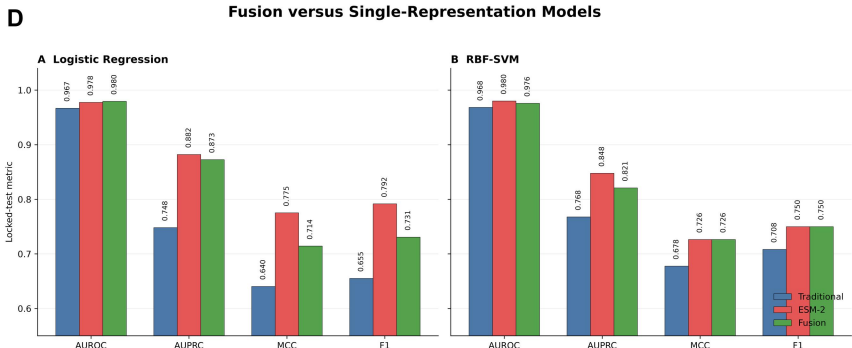


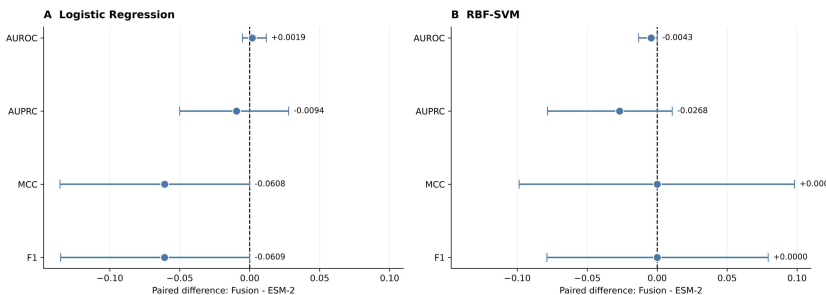


Each fold independently fitted scalars, >=90% PCA dimensions, and CCA on 576 training peptides; error bars show validation SD across five folds.



Point estimates on the frozen 181-peptide test set; superiority is not inferred without paired uncertainty analysis.

E Paired Bootstrap Comparison of Fusion and ESM-2-Only Models



Positive values favor Fusion. Diamonds indicate intervals excluding zero; circles indicate intervals including zero. No multiplicity-adjusted claim is made.

F Stability of ESM-2 latent-dimension Importance across development CV folds

